



Full Stack Assignment



Objective

The goal of this assignment is to evaluate your ability to design, build, and deliver a **production-grade full-stack application**.

We are assessing:

- System design thinking
- Code quality and maintainability
- API design and backend architecture
- Frontend engineering and UX
- Database design and performance
- Scalability and edge-case handling
- Ownership and decision-making



Problem Statement

Build a “**Contract Management Dashboard**” (or similar domain-based system).

The application should allow users to:

- Create and manage contracts
- Track contract lifecycle stages
- Search and filter contracts
- View contract details and version history



Functional Requirements

1. Authentication & Authorization

- User signup/login (JWT/session-based)
- Role-based access:
 - Admin
 - User

2. Contract Management

Create Contract

- Title
- Description
- Parties involved
- Start date / End date
- Status (Draft, Active, Executed, Expired)

Update Contract

- Editable fields
- Status transitions with validation

Delete Contract

- Soft delete (**Must**)

3. Contract Listing Page

- Search by:
 - Title
 - Party name
- Filters:
 - Status
 - Date range
- Sorting:
 - Created date
 - Updated date

4. Contract Detail Page

- Show complete contract information
- Activity timeline (audit logs): **Optional**
- Version history (basic implementation acceptable)

5. Version History (Optional)

- Each update should create a version snapshot
- Ability to:
 - View previous versions
 - Compare changes (basic diff is fine)

6. API Requirements

- RESTful APIs
- Proper status codes
- Validation and error handling
- Pagination and filtering support



Frontend Requirements

- Framework: React.js (Javascript, no typescript)
- Responsive UI
- Clean UX (focus on usability, not heavy design)
- State management (Redux)



Backend Requirements

- Node.js (preferred)
- Input validation
- Error handling middleware



Database Design

- Use MongoDB / PostgreSQL
- Proper schema design
- Relations or embedding (justify choice)